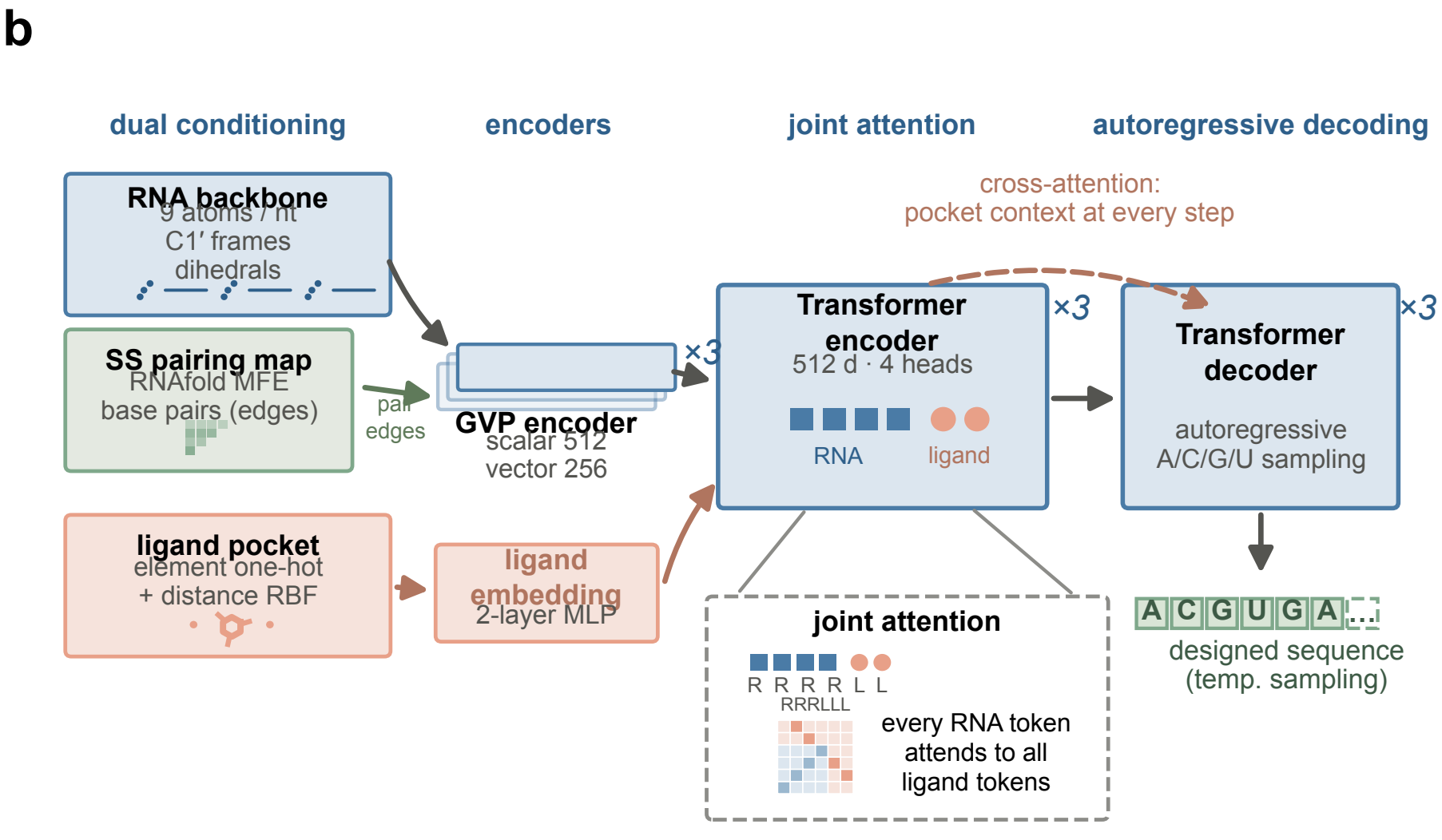
b
dual conditioning
encoders
joint attention
autoregressive decoding
RNA backbone
9 atoms / nt
C1′ frames
dihedrals
SS pairing map
RNAfold MFE
base pairs (edges)
pair edges
GVP encoder
scalar 512
vector 256
×3
ligand pocket
element one-hot
+ distance RBF
ligand embedding
2-layer MLP
cross-attention:
pocket context at every step
Transformer encoder
512 d · 4 heads
RNA
ligand
×3
Transformer decoder
autoregressive
A/C/G/U sampling
×3
A C G U G A …
designed sequence
(temp. sampling)
joint attention
R R R R L L
RRRLLL
every RNA token
attends to all
ligand tokens